Assignment 2

In this assignment, you will create both the **front-end** and the **back-end** of a web application. This application must display a **collection** of some sort, but the **data is up to you**. The collection must also include a way to **filter** the items using **categories of your choice**.

All **data** in this assignment should be stored in a **database**.

Express Requirements:

1. **Router:** Organize your routes using **Express Router**
2. **CRUD Operations:** Implement **at least one** endpoint for **each** of the **Create**, **Update**, and **Delete** operations. **You will need at least two endpoints for Read.** One to display **all the entries**, and another to display the data from a **single entry**
3. **File Upload:** Allow users to **upload files** to a static folder that is accessible in the react application
4. **Middleware:**
    - Use `mysql2` for database interactions, `multer` for handling file uploads and `cors()` to support requests from the front-end React application

Database Requirements:

1. **Two Tables:** The application must use **at least two** tables
    - **Primary Table:** This table will store the **main entries** that are displayed in your application including the **file name** for any **uploaded images**
    - **Secondary Table**: This table will store some kind of **meta data** like **categories** or **sizes** that will be referenced via **foreign key** on the **Primary Table**
    - **Note:** Tables should be named with context of the information they hold, **not** primary_table and secondary_table
    - **Note:** Make sure each column in the table is set to the appropriate data type

React Requirements:

1. **All Items:** Create a **component** that displays a short introduction to **all items** in the database (archive)

2. **Single Item:** Create a **component/page** that displays all the information related to a **single entry**
3. **Filtering:** Add a component to the listing page that allows a user to **filter the items** based on their **category** or **size** etc., you are only required to filter by one column
4. **Add Item:** Create a **form component** to **add a new item** to the database
5. **Remove Item:** Create a **UI element** that allows users to **delete an item** from the database
6. **Edit Item:** Create a **form component** to **edit** an **existing item** in the database

Deliverables:

- A single **.zip file** containing both the **front-end** and **back-end** of your application
- **node_modules** must be **removed**
- Include **detailed comments** on the files you create **in your own words** (not copied from an LLM or in-class assignments)
- Include an **exported .sql file** in your **api folder, I** am happy to help if you need assistance with the export
- Please 🙏 name your database to include your initials to make it **unique**

Example Project Directory Structure

```
assignment-2/
├── api/
│   └── express application
│   └── exported-db.sql
├── web/
│   └── react application
```

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Due March 23 at 11:59 PM